

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
k = 5
model = KNeighborsClassifier(n_neighbors=k)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("K-Nearest Neighbours (K-NN)")
print("Value of K:", k)
print("Accuracy:", accuracy)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)
```

```
print("\nNew Sample:", new_sample)
print("Predicted Class:", iris.target_names[prediction[0]])
```

OUTPUT:

```
... K-Nearest Neighbours (K-NN)
Value of K: 5
Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

New Sample: [[5.1, 3.5, 1.4, 0.2]]
Predicted Class: setosa
```